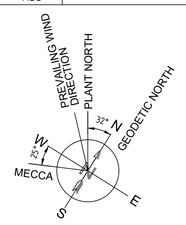
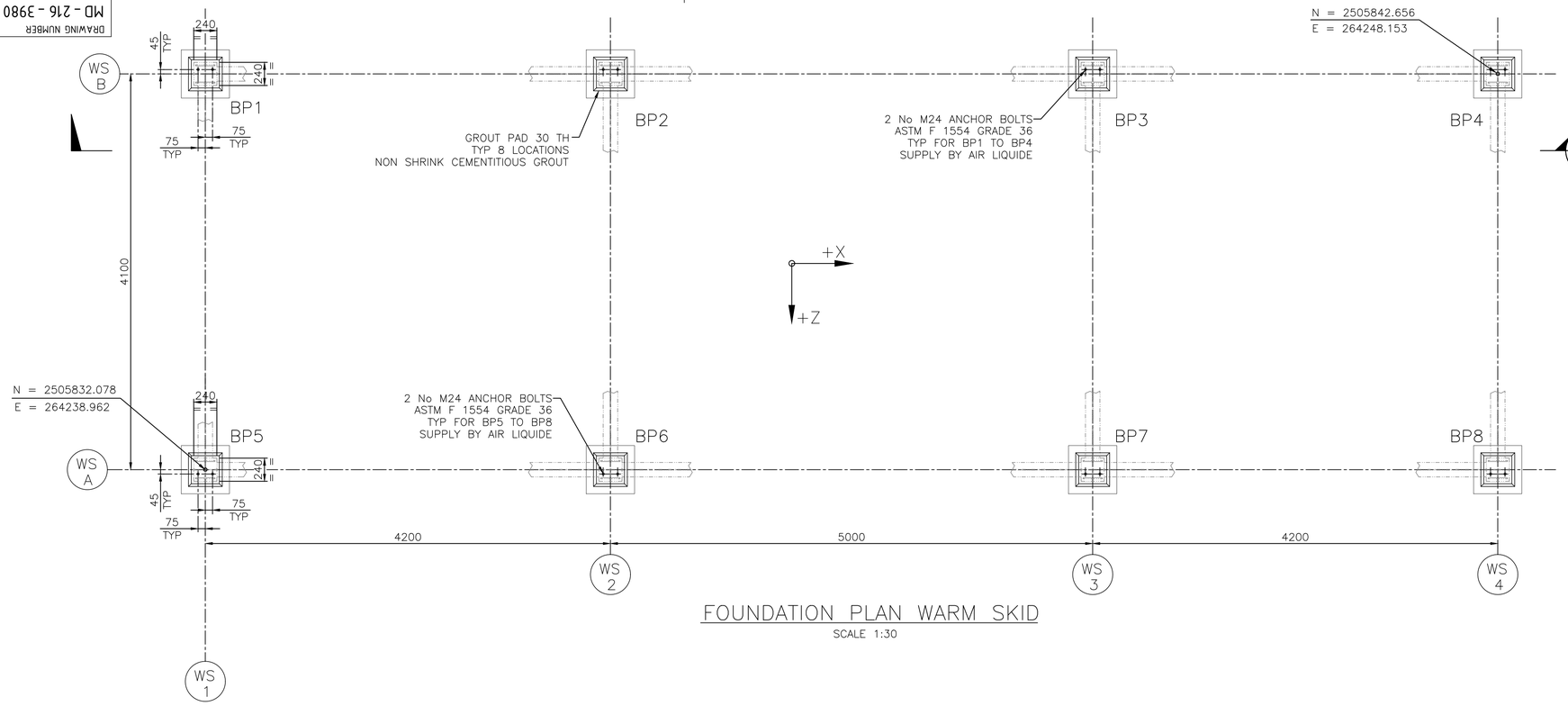
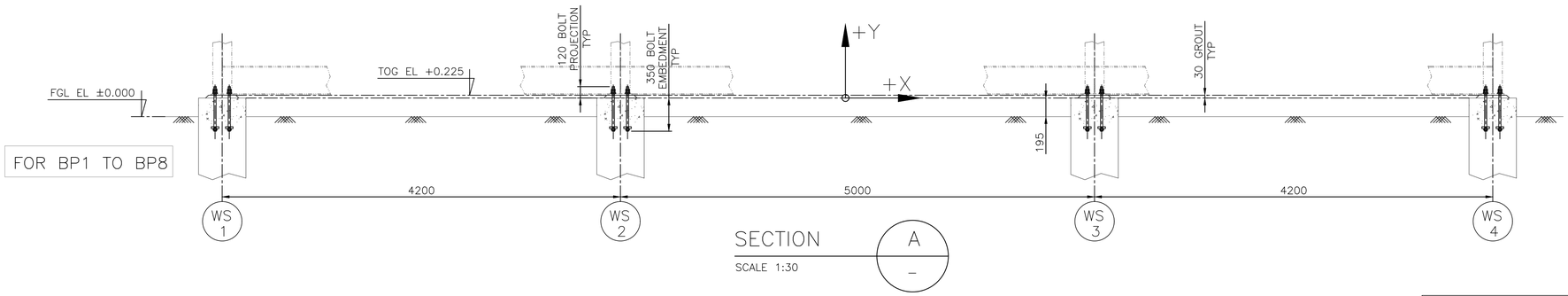
DRAWING NUMBER: MD - 216 - 3980 - EG - CV - GAD - V551-1478 | REV: D01





FOUNDATION PLAN WARM SKID

SCALE 1:30



FOR BP1 TO BP8

SECTION A – A

SCALE 1:30

NOTES

1. ALL DIMENSIONS ARE IN MILLIMETER.
2. ALL LEVELS ARE IN METER.
3. FGL LEVEL ±0.0 CORRESPONDS TO RL 912.350.
4. THE LOADS INDICATED IN LOAD DATA TABLE ARE ACTUAL. FOUNDATION DESIGNER SHOULD CONSIDER NECESSARY MARGIN.
5. SIZE OF FOUNDATION AND PEDESTAL SHALL BE DECIDED BY CIVIL DESIGN ENGINEER AS PER GIVEN LOADING.
6. ADDITIONAL REBAR REQUIREMENT AROUND ANCHOR BOLTS TO ENSURE NO FAILURE OF CONCRETE SHALL BE DECIDED BY CIVIL DESIGN ENGINEER AS PER GIVEN LOADING.
7. WIND & SEISMIC LOADS ARE AS PER SBC 301.

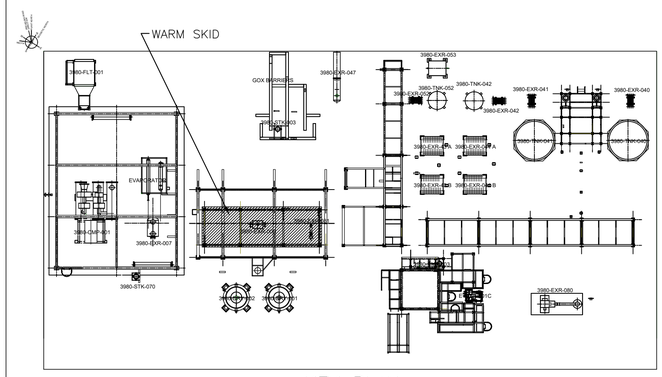
LEGEND :

CL – CENTER LINE
CG – CENTER OF GRAVITY
CONC – CONCRETE
EL – ELEVATION
EQP – EQUIPMENT
FGL – FINISHED GROUND LEVEL
Gr – GRADE
MPa – MEGA PASCAL
mm – MILLIMETER
No – NUMBER / NUMBERS
SQ – SQUARE
TH – THICKNESS
TOC – TOP OF CONCRETE
TOG – TOP OF GROUT
TOP – TOP OF SHIM PLATE
TYP – TYPICAL

FOUNDATION LOAD : WARM SKID

SL No	Description	BP1			BP2			BP3			BP4			BP5			BP6			BP7			BP8			Remarks
		X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	
1	DEAD LOAD	0.0	-10.1	-0.5	0.0	-19.2	-0.8	0.0	-22.8	-1.6	0.0	-12.8	-0.7	0.0	-10.3	0.5	0.0	-17.8	0.8	0.0	-21.5	1.6	0.0	-11.8	0.7	A. ALL LOADS FURNISHED IN THIS TABLE ARE UNFACTORED ACTING LOADS. B. PIPE LOAD AND EQUIPMENT LOAD SHALL BE CONSIDERED UNDER DEAD LOAD CATEGORY. C. FOR REVERSE DIRECTION OF LOAD CASE 6 TO 11, MULTIPLY ALL VALUES WITH -1.
2	EQUIPMENT LOAD	0.0	-18.9	-1.0	0.0	-2.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	-7.4	1.0	0.0	-1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
3	PIPE EMPTY LOAD	0.0	-18.2	-0.8	0.0	-51.2	-2.0	0.0	-55.0	-5.2	0.0	-23.8	-1.3	0.0	-12.7	0.8	0.0	-52.8	2.0	0.0	-57.2	5.2	0.0	-18.6	1.3	
4	PIPE OPERATING LOAD	0.0	-28.4	-1.1	0.0	-88.8	-3.3	0.0	-93.1	-7.5	0.0	-39.1	-1.8	0.0	-19.1	1.1	0.0	-78.4	3.3	0.0	-86.0	7.5	0.0	-25.9	1.8	
5	LIVE LOAD	0.0	-24.5	-1.1	0.0	-53.8	-2.8	0.0	-65.8	-6.0	0.0	-27.5	-2.1	0.0	-24.7	1.1	0.0	-54.9	2.8	0.0	-67.1	6.0	0.0	-26.4	2.1	
6	PIPE LOAD +X	4.9	0.8	-0.5	35.0	8.7	-0.7	21.1	0.3	0.9	9.9	-9.7	1.4	9.1	4.2	-2.6	41.8	-0.8	-7.0	42.0	-2.0	6.0	1.6	-1.4	2.4	
7	PIPE LOAD +Z	0.5	-3.3	-5.4	2.9	-2.7	-9.2	1.4	7.1	-10.7	-0.3	8.3	4.7	1.7	3.5	-9.8	-1.9	2.6	-29.1	-4.2	-7.2	-20.7	0.0	-8.2	4.2	
8	WIND LOAD +X	3.6	-0.4	-0.2	6.6	0.4	-0.2	4.1	15.1	0.3	18.9	-15.0	0.3	18.7	14.5	-0.6	6.1	-14.6	-0.8	7.1	0.2	0.8	2.0	0.0	0.3	
9	WIND LOAD +Z	0.2	5.8	7.3	-0.4	8.7	10.7	-0.3	14.0	15.4	0.3	6.5	8.0	-0.5	-6.0	8.5	0.3	-8.6	11.5	0.5	-13.8	12.3	0.0	-6.8	7.0	
10	SEISMIC LOAD +X	0.3	0.0	0.0	0.9	0.2	0.0	0.6	5.9	0.2	7.4	-6.0	0.0	5.5	4.8	-0.2	0.5	-4.9	-0.2	0.7	0.2	0.0	0.5	0.0	0.0	
11	SEISMIC LOAD +Z	0.2	2.4	1.5	0.0	1.7	1.5	-0.2	5.0	4.4	0.0	2.5	1.9	-0.2	-2.4	1.4	0.0	-1.7	1.5	0.2	-5.0	3.0	0.0	-2.6	1.9	

NOTE: PIPE LOAD +X & PIPE LOAD+Z ARE THE HORIZONTAL LOADS GENERATED DUE TO PIPE STRESS.
THESE TWO LOADS SHALL BE CONSIDERED ACTING SIMULTANEOUSLY IN OPERATING CONDITION.



KEY PLAN

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		D01	22/02/2023	AS-BUILT	--	STY/SYC/SHH	--	BDA/PRI/JPD	
		C01	18/08/2020	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER'S COMMENT ON 14/07/2020	--	STY/SYC/SHH	--	BDA/PRI/JPD	
		A01	30/05/2020	ISSUED FOR INFORMATION	--	STY/SYC/SHH	--	BDA/PRI/JPD	

Air Liquide ENGINEERING & CONSTRUCTION

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DRAWING NUMBER		REVISION	STATUS
12126-60-01-CS-141030		0	AFD
DATE 11/05/2020	DATE 12/05/2020	DATE 30/05/2020	
DRAWN SGH	CHECKED ABM	APPROVED DS	

LARSEN & TOUBRO DRAWING NUMBER	REVISION
019045_080_C_V_24225_005	2
OUTOTEC DRAWING NUMBER	DOCUMENT ID / REVISION
900707 __GAU01 __AG22__MTL01 __39818	OU100680043 / 2

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DRAWN	--	--
CHECKED	STY/ SYC/ SHH	28/05/2020
ENGINEER	--	--
APPROVED	BDA/ PRI/ JPD	30/05/2020
EMPLOYER		
SCALE	1:30	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
OXYGEN PLANT
FOUNDATION LOAD & ANCHORAGE - WARM SKID

DRAWING NUMBER: MD - 216 - 3980 - EG - CV - GAD - V551-1478 | REV: D01